# Sang Hak Lee

Marine Data Engineer · Vessel & weather data collection, datasets and quality

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Data engineer who **collects and cleans** AIS / V-PASS vessel data and marine-weather forecast data (NetCDF), and **builds datasets** on PostgreSQL data models I design. Embeds spec-based **data quality validation** and processing **metadata management** into pipelines; licensed weather forecaster with meteorology and ocean-environment certifications, bringing domain understanding of vessel, weather and route data.

## EXPERIENCE

### GeoSR (Geo System Research) · Data Engineer

Feb 2023 - Present

- **Vessel data collection & cleansing** — Cleaned TB-scale AIS trajectories for the Busan area (outliers, duplicates, time alignment) and joined route/anchorage data to build anchoring, loitering and route-deviation candidate datasets (GeoPandas, multiprocessing).
- **V-PASS dataset construction** — Streaming merge and filtering of vessel observations (data years 2023-2025) into regional/grid statistical datasets and density maps, joined with catch-grade data.
- **Weather & ocean data pipeline** — Automated NetCDF/CSV ingestion → spec validation → raster and current-vector conversion → PostgreSQL loading → GeoServer publishing (Python, Shell, Docker, air-gapped).
- **Data modeling** — Split loading for 11 monitoring/forecast products from 4 institutions into a **file catalog** (data type, year-month, monitoring/forecast, publication flag), **validation results** (per-file pass/fail and error items) and **value tables** (current lat/lon, speed, direction, depth), linked by file name.
- **Data quality validation** — Automated spec checks (Class, Variable, Level, Unit) with error records, exclusion and count reconciliation, service-point to model-grid matching, old vs. new product comparisons.
- **Metadata management** — Maintained processing/error history, input spec sheets and data specifications; traced source file → validation result → loaded values per file; SHA-256 change detection re-validates only changed files.
- **Systems & documentation** — Built forecast production servers (Oracle DB, Tomcat), scheduling, monitoring, incident response; requirements and test documents; conference poster (2024), workshop talk (2025).

### Korea Meteorological Industry Promotion Agency · R&D Planning Intern

Jun - Sep 2022

- Researched weather technology and policy trends; wrote monthly newsletters and quarterly reports.

## SELECTED PROJECTS

### Marine Travel AI Agent · GeoSR internal R&D

2026

Managed public data consumed by an LLM agent — automated TourAPI and 8 marine-forecast feed collection (n8n), duplicate-ID, coordinate-range, missing-page and per-feed freshness checks, collection/issue timestamps, typed validation tools for LLM use (LangGraph, FastAPI).

### AIS Trajectory Analytics

2024 - 2025

Activity-candidate detection from AIS tracks with route and anchorage polygons; KOSMEE autumn conference poster (2024) and workshop talk (2025).

### Ocean Forecast GeoPipeline

2025.06 - 2026.04

NetCDF/CSV spec validation → spatial conversion → PostgreSQL loading → GeoServer publishing; code and validation docs on GitHub.

## SKILLS

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|---|---|
| **Data** | Python, SQL, PostgreSQL, Oracle, Pandas, GeoPandas |
| **Formats** | NetCDF, AIS, V-PASS, GeoTIFF, GeoServer, QGIS |
| **Infra** | Linux, Shell, Docker, scheduling & monitoring |
| **Other** | Fortran, Rust, LangGraph, FastAPI, Ollama (LLM agents) |

## CREDENTIALS · EDUCATION

Weather Forecaster License · Engineer Meteorology · Engineer Ocean Environment

B.S. Earth & Environmental Sciences, Jeonbuk National University (2016 - 2022)

Research Club Grand Prize, JBNU College of Natural Sciences (2021)

## ① Monitoring & forecast product loading pipeline

GeoSR · Jun 2025 - Apr 2026

Air-gapped pipeline that validates 11 monthly marine/weather monitoring and forecast products (NetCDF, CSV) from 4 institutions (KIOST, NIFS, KHOA, KOPRI), loads them into PostgreSQL and publishes GeoServer layers.

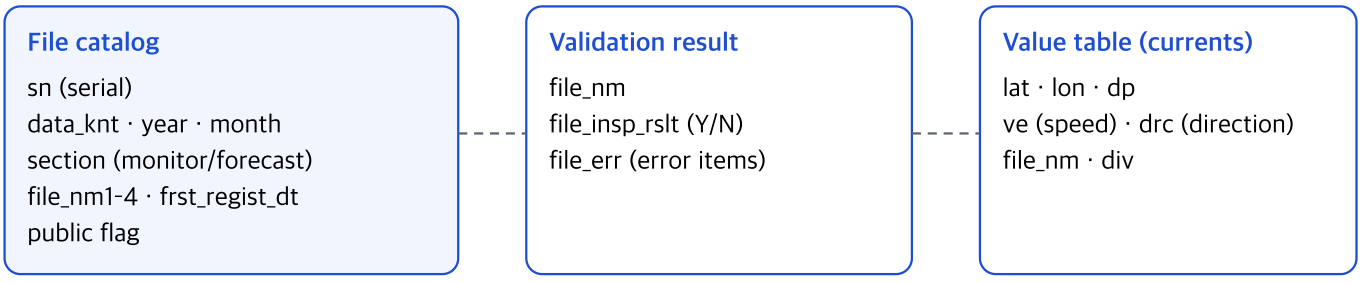
**Role** Data model design, spec validation, spatial conversion and ETL; specifications and test documents

**Design** Separated product **metadata** (data type, year-month, monitoring/forecast, registration time, publication flag), **validation results** and **values** into distinct tables; the catalog is queried and updated by (data type, year, month, section), while results and values link by file name to trace source → validation → load

**Validation** Automatically compared Class, Variable, Level (depth) and Unit against the input spec; recorded pass/fail and error items per file

**Processing** File watching (Python, Shell, Docker), raster regridding/reprojection, speed and direction from U/V, upserts, transactions, reprocessing, scheduling

**Results** Item-level reports, GeoTIFF and current-vector CSV, GeoServer layers; code and validation docs on GitHub




Loading structure (summary of actual tables)

**Change detection & incremental runs** Kept a processing log of size, mtime and SHA-256 per file (filename, size, mtime, sha256, processed_at) and re-ran spec validation only for changed files.

## ② AIS vessel data cleansing & activity detection

GeoSR · 2024 - 2025

Combined historical Busan-area AIS with vessel metadata and route/anchorage polygons to detect and visualize anchoring, loitering and route-deviation candidates.

**Role** Trajectory preprocessing, spatial-rule and time-alignment detection, parallel visualization; poster and workshop talk

**Cleansing** Removed outliers and duplicates from TB-scale raw data, aligned tracks per vessel in time, derived speed, heading and displacement

**Design** Judged candidates with **readable rules** combining polygon membership and movement features so each decision can be reviewed

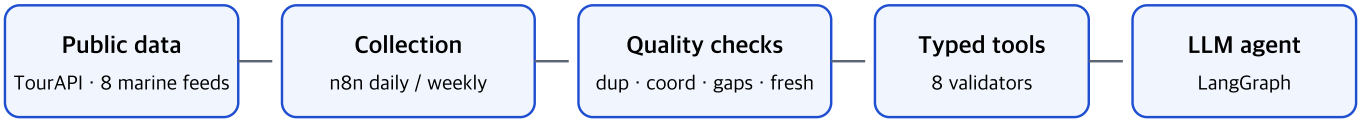
**Processing** GeoPandas spatial joins; multiprocessing for large map time-lapse generation

**Results** KOSMEE 2024 autumn conference poster, Jan 2025 workshop talk, code public on GitHub

## ③ Marine Travel AI Agent — public data for an LLM

GeoSR internal R&D · 2026

Collects and validates tourism, marine-forecast, weather and transport public data so an LLM agent can cite checkable evidence. Quality management of **external data an LLM consumes**, not training-dataset construction.




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|---|---|
| **Role** | Public-data integration, collection workflows, quality checks and typed validation tools, itinerary rule engine |
| **Quality** | Content-ID duplicates, coordinate range, missing/repeated pages, per-feed freshness; atomic refresh keeps prior data on failure |
| **Metadata** | Stored collection time, forecast issue time and source with each record so the time basis of an LLM answer can be checked |
| **Scale & tests** | ~10K tourism records, 8 marine forecast feeds; 195 Python and 28 Node tests passing; Langfuse execution tracing |

## ④ V-PASS vessel observation datasets

GeoSR · data years 2023 - 2025

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|---|---|
| **Processing** | Annual streaming merge, speed and vessel-ID filters, monthly/seasonal aggregation, regional grid assignment and count sums, weighted Quartic KDE, catch-grade join |
| **Validation** | Exclusion, grid-assignment and count reconciliation reports matching input and output totals; methodology and validation records public |

## ⑤ Forecast system build, operation & analysis

GeoSR · 2023 - 2025

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|---|---|
| **Build** | Forecast-index and sea-condition chart production servers: quotes, install, remote access, packet filtering, Oracle DB, Tomcat, Python/Java; backup NAS (2025) |
| **Operation** | Production scheduling, daily output monitoring, incident response and reporting |
| **Analysis** | Old vs. new model product comparisons, severe-weather-day and website/OpenAPI usage analysis, monthly statistics and summaries |